

SHEARWATER, NS, Canada

WMO: 716010

Lat: 44.6297N

Lon: 63.5133W

Elev: 144

StdP: 14.62

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.8	8.1	-5.9	4.0	6.2	-1.7	5.0	10.3	31.2	32.0	28.3	31.6	10.9	310	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	79.7	67.4	76.9	65.9	74.3	64.8	70.4	76.0	68.8	73.6	67.2	71.5	9.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	105.2	72.7	67.0	99.7	71.0	65.5	94.8	69.5	34.4	76.5	33.0	73.9	31.8	71.8	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	22.3	19.7	-1.7	86.8	5.6	3.2	-5.7	89.1	-9.0	91.0	-12.1	92.8	-16.1	95.1		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			-2.7	73.9	5.3	1.7	-6.5	75.1	-9.6	76.1	-12.6	77.0	-16.4	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	25.4	26.1	31.8	40.5	49.2	57.9	65.4	66.2	60.7	50.8	41.3	32.1
	DBStd	15.78	9.63	8.53	7.41	5.67	5.36	5.71	4.20	3.96	5.58	6.41	7.50	8.48
	HDD50	3269	762	669	565	290	79	5	0	0	1	68	274	556
	HDD65	7192	1227	1089	1029	736	490	223	45	31	150	440	711	1021
	CDD50	1714	0	0	2	4	55	241	478	504	323	94	13	0
	CDD65	160	0	0	0	0	1	9	58	70	21	1	0	0
	CDH74	651	0	0	2	0	16	80	259	226	66	2	0	0
	CDH80	95	0	0	0	0	2	15	41	25	12	0	0	0
Wind	WSAvg	9.2	10.9	10.7	10.7	10.2	8.7	7.8	6.8	6.9	7.9	9.3	10.1	10.4
Precipitation	PrecAvg	55.7	5.2	4.5	4.8	4.3	4.4	4.1	4.0	3.8	3.8	5.0	5.5	5.9
	PrecMax	70.7	10.7	9.0	9.6	8.2	8.4	13.7	9.0	12.6	8.9	10.1	10.2	10.0
	PrecMin	38.7	1.7	1.3	1.8	0.7	1.0	1.3	0.5	1.0	1.1	1.3	1.5	2.6
	PrecStd	7.8	1.9	1.9	1.8	1.7	1.8	2.2	1.9	2.4	1.8	2.0	2.2	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	48.3	56.9	62.9	75.7	82.1	85.0	83.3	79.9	71.1	60.6	54.2
		MCWB	50.2	46.0	48.4	50.1	60.1	66.0	70.1	68.6	68.6	61.3	57.6	52.7
	2%	DB	46.9	44.2	48.2	56.8	66.9	75.4	79.5	79.0	75.0	66.1	57.8	51.0
		MCWB	45.8	42.3	43.0	47.0	55.0	63.1	67.7	67.3	65.8	60.9	55.6	49.7
	5%	DB	43.2	40.8	44.8	52.7	62.0	71.7	76.6	76.4	71.7	63.2	55.5	48.0
		MCWB	41.8	39.2	41.2	45.5	52.6	61.6	66.1	66.1	64.5	59.2	53.7	46.8
	10%	DB	39.1	38.0	41.9	49.4	58.2	68.0	73.8	73.7	69.0	60.8	53.0	44.8
		MCWB	37.6	36.1	38.8	44.3	50.4	59.7	64.9	65.2	63.5	57.3	51.0	43.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	46.7	49.8	53.6	62.5	69.8	72.2	72.0	71.7	65.5	59.1	53.5
		MCDB	51.3	47.7	52.7	57.8	70.1	77.9	80.2	78.3	76.2	68.0	60.1	54.1
	2%	WB	46.1	42.8	44.8	49.9	57.5	66.1	70.2	70.1	68.9	62.8	56.5	50.0
		MCDB	46.7	43.8	46.5	53.3	63.7	72.0	76.8	74.9	72.3	64.9	57.3	50.7
	5%	WB	42.0	39.5	42.1	47.6	54.5	63.5	68.4	68.6	66.7	60.4	54.1	47.0
		MCDB	42.9	40.5	44.0	50.5	59.2	68.4	73.7	73.2	69.7	62.3	55.3	47.8
	10%	WB	37.8	36.4	39.1	45.2	52.0	61.3	66.7	67.2	65.1	58.0	51.3	43.3
		MCDB	39.0	37.6	41.3	48.7	56.3	65.9	70.9	71.5	67.9	60.2	52.7	44.5
Mean Daily Temperature Range	5% DB	MDBR	13.7	13.4	12.8	13.2	14.2	14.5	13.8	13.8	13.9	12.6	11.5	12.1
		MCDBR	17.7	15.1	16.4	19.3	21.3	21.2	18.9	17.7	16.2	14.5	13.3	15.4
		MCWBR	18.3	14.8	13.3	12.9	12.9	12.2	9.7	9.3	10.1	10.9	13.1	16.2
	5% WB	MCDBR	18.0	15.1	15.3	15.9	17.8	18.2	15.6	14.1	12.7	12.8	12.9	15.0
		MCWBR	18.8	15.4	14.3	12.8	12.7	12.2	9.6	8.9	9.3	11.6	13.5	16.2
Clear-Sky Solar Irradiance	taub	0.302	0.296	0.323	0.348	0.369	0.398	0.414	0.395	0.353	0.332	0.315	0.296	
	taud	2.473	2.521	2.476	2.449	2.437	2.381	2.358	2.409	2.500	2.555	2.552	2.537	
	Ebn at Noon	261	286	291	290	286	277	270	272	277	269	255	253	
	Edh at Noon	22	25	30	34	35	37	38	35	29	24	21	19	
All-Sky Solar Radiation	RadAvg	455	740	1119	1367	1570	1707	1696	1584	1291	837	481	359	
	RadStd	35	54	76	112	202	159	124	110	85	87	52	28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.67	+0.90	N/A	N/A	+101	+31

Nomenclature: See separate page